MATH317 · NUMERICAL METHODS

# Chapter 3.2 Newton's Method

Midterm summary · Lecture 6

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## 01 What is Newton's method?

A method for finding a root of $f(x) = 0$. It is also called the **Newton-Raphson iteration**. It is one of the most important procedures in numerical analysis and is much faster than bisection.

**Requirement**

The function must be **Differentiable**, because the method needs $f'(x)$.

### The formula

$$x_{n+1} = x_n - f(x_n) / f'(x_n)$$

Start from a guess $x_0$ and substitute to get $x_1$. Substitute $x_1$ to get $x_2$, and so on.

## 02 Where does the formula come from?

### First view: the tangent line

At the point $(x_0, f(x_0))$ draw the tangent. Near that point the tangent is close to the curve, and its equation is:

$$l(x) = f'(x_0)(x - x_0) + f(x_0)$$

Instead of the root of the hard function, find the root of the tangent. Set $l(x) = 0$ and solve:

$$x_1 = x_0 - f(x_0) / f'(x_0)$$

### Second view: Taylor series

Ask: what correction `h` added to `x₀` gives the root? We want `f(x₀ + h) = 0`.

$$f(x_0 + h) = f(x_0) + h\, f'(x_0) + h^2/2!\, f''(x_0) + \ldots = 0$$

Keep the first two terms only: $f(x_0) + h\, f'(x_0) = 0$

$$h = -\, f(x_0) \,/\, f'(x_0)$$

$$x_1 = x_0 + h = x_0 - f(x_0) \,/\, f'(x_0)$$

Both views give the same formula.

## 03 The slide example

If $f(x) = x^3 - x + 1$ and $x_0 = 1$, what are $x_1$ and $x_2$ in the Newton iteration?

#### 1 Differentiate

$$f(x) = x^3 - x + 1 \qquad f'(x) = 3x^2 - 1$$

#### 2 Compute `x₁`

$$f(1) = 1 - 1 + 1 = 1 \qquad f'(1) = 3 - 1 = 2$$

$$x_1 = 1 - 1/2 = 1/2$$

### 3 Compute $x_2$

$$f(1/2) = 1/8 - 1/2 + 1 = 5/8 \qquad f'(1/2) = 3/4 - 1 = -1/4$$

$$x_2 = 1/2 - (5/8) \,/\, (-1/4) = 1/2 + 5/2$$

$$x_2 = 3$$

**Notice**

Here $x_2 = 3$ jumped far from $x_0$. Newton's method does not guarantee convergence when the first guess is far from the root.

## 04 An example showing Newton's speed

The slides solve $x^3 + x = 2x^2 + 3$. Move everything to one side:

$$f(x) = x^3 - 2x^2 + x - 3 = ((x - 2)x + 1)x - 3$$

$$f'(x) = 3x^2 - 4x + 1 = (3x - 4)x + 1$$

Both are written in **Nested form** so they are cheaper to evaluate. Start from $x_0 = 3$:

| n | $x_n$ | $f(x_n)$ |
|---|---|---|
| 0 | 3.0 | 9.0 |
| 1 | 2.4375 | 2.04 |
| 2 | 2.21303 | 0.256 |
| 3 | 2.17555 | $6.46 \times 10^{-3}$ |
| 4 | 2.17456 | $4.48 \times 10^{-6}$ |
| 5 | 2.17456 | $2.2 \times 10^{-12}$ |

The real root is `2.17456`. Watch how fast `f(xₙ)` shrinks: the number of correct digits roughly doubles every step.

The slide shows this table as an image, so the numbers here were computed with the same formula.

## 05 Convergence analysis

Let the error at step n be `eₙ = xₙ − r`, where `r` is the root.

**Theorem 1**

The errors of Newton's method are related by:

$$e_{n+1} = -\tfrac{1}{2} \cdot f''(\xi_n) \,/\, f'(x_n) \cdot e_n^2 \qquad (\xi_n \text{ between } x_n \text{ and } r)$$

$$|\, e_{n+1} \,| \;\le\; c \cdot |\, e_n \,|^2$$

The new error is proportional to the square of the old one. This is called **Quadratic convergence**.

| If the error is now | After one step it is about |
|---|---|
| $10^{-2}$ | $10^{-4}$ |
| $10^{-4}$ | $10^{-8}$ |

**Condition**

This holds only when $x_0$ is sufficiently close to the root.

## 06 When do we stop?

After each step compute the **Absolute relative approximate error**:

$$|\varepsilon_a| = |(x_{i+1} - x_i) / x_{i+1}| \times 100$$

**1** **Compute $|\varepsilon_a|$.**

**2** **Compare it with the given tolerance $\varepsilon$.**

**3** **If $|\varepsilon_a| > \varepsilon$ continue. Otherwise stop.**

Also stop when the number of iterations exceeds the allowed maximum.

### The pseudocode on the slide

It starts from a value `x` and runs at most `n_max` iterations. It needs two functions, `f(x)` and `f'(x)`. The parameters $\varepsilon$ and $\delta$ control convergence and accuracy.

## 07 Newton vs Bisection

| | Bisection | Newton |
|---|---|---|
| What it needs to start | An interval `[a, b]` with a sign change | One guess $x_0$ |
| Needs the derivative? | No | Yes |
| Speed | Slow: the error halves each step | Fast: **Quadratic convergence** |
| Is convergence guaranteed? | Yes | No, unless $x_0$ is close to the root |
| Work per step | One evaluation: `f(c)` | Two evaluations: `f` and `f'` |

This table is my arrangement of facts stated in Lectures 5 and 6.

## 08 How it shows up in the midterm

Newton's method appeared in one of the four midterms (461), for five marks.

> Given $f(x) = x^3 - 2x - 1$ and $x_0 = 1.5$, apply Newton's method for two iterations.

$$f'(x) = 3x^2 - 2$$

$$f(1.5) = 3.375 - 3 - 1 = -0.625 \qquad f'(1.5) = 6.75 - 2 = 4.75$$

$$x_1 = 1.5 - (-0.625)/4.75 = 1.5 + 0.1316$$

$$\mathbf{x_1 = 1.6316}$$

$$f(1.6316) = 0.0803 \qquad f'(1.6316) = 5.9864$$

$$x_2 = 1.6316 - 0.0803/5.9864 = 1.6316 - 0.0134$$

$$\mathbf{x_2 = 1.6182}$$

If the question also asks for the error between the last two values:

$$|\varepsilon_a| = |(1.6182 - 1.6316) / 1.6182| \times 100 = 0.83\ \%$$

The $|\varepsilon_a|$ line is my addition; the midterm did not ask for it. Past papers show the style of the question, not a promise about the next exam.

## 09 Common mistakes

| Mistake | Fix |
| --- | --- |
| Adding instead of subtracting in the formula | $x_{n+1} = x_n - f/f'$. Watch the sign when $f$ is negative. |
| Differentiating incorrectly | Write $f'(x)$ on the first line and check it. |
| Substituting $x_0$ again in the second step | The second step uses $x_1$. |
| Rounding $x_1$ too much before continuing | Keep at least four decimal places. |
| Saying Newton always converges | No. It needs $x_0$ close to the root. |

## 10 Check yourself

1. $f(x) = x^2 - 2$ and $x_0 = 1$. Find $x_1$ and $x_2$.
2. $f(x) = x^3 - x - 1$ and $x_0 = 1.5$. Find $x_1$ and $x_2$.
3. What does quadratic convergence mean?

## Solutions

1) $f'(x) = 2x$

$x_1 = 1 - (1 - 2)/2 = 1 + 0.5 = 1.5$

$x_2 = 1.5 - (2.25 - 2)/3 = 1.5 - 0.0833$

**$x_2 = 1.4167$** (true root $\sqrt{2} = 1.4142$)

2) $f'(x) = 3x^2 - 1$

$f(1.5) = 0.875$ $f'(1.5) = 5.75$

$x_1 = 1.5 - 0.875/5.75 = 1.3478$

$f(1.3478) = 0.1007$ $f'(1.3478) = 4.4499$

**$x_2 = 1.3478 - 0.1007/4.4499 = 1.3252$**

**3)** The new error is proportional to the square of the old one: $|e_{n+1}| \le c\,|e_n|^2$. The number of correct digits roughly doubles each step.

The check-yourself questions are original practice items, not past-exam questions.